Where the Signal Is: Label Provenance, Encoder Choice and the Limits of Fusion in Arabic Multimodal Affect Recognition

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Code, data and documents: `https://github.com/nourgaser/giu-ms-rp`

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Abstract

Cross-modal attention is the dominant architectural response to multimodal affect recognition. We set out to test whether it outperforms late fusion for Arabic, and in particular whether its advantage concentrates on samples where the modalities carry conflicting evidence. Across two corpora, twelve fusion arrangements and four encoder families, it does not — and the reason is instructive. On Ar-MuSA, the only openly available Arabic multimodal sentiment corpus, we find that the overall label agrees with the corpus's own transcript-level annotation on **96.5%** of samples, so a text-only model already holds nearly all the label information and no fusion mechanism has room to distinguish itself. We predicted from this that a corpus whose label is *not* recoverable from text would behave differently, and tested that prediction on EGY-MER, an Egyptian Arabic emotion corpus obtained during this work. The prediction holds: on EGY-MER the same measurement gives **59.4%** for transcript against **91.1%** for audio, the ordering of the modalities inverts, and the multimodal gain over the best single modality triples. Yet even there, no attention arrangement we tried beats plain concatenation by more than the variation between seeds, and concatenation is the smallest model by a factor of two to seven. Every measurable improvement we obtained came from representation — dialect-matched text encoders (+0.054), reading audio from mid-stack rather than the final layer (+0.053), expression-pretrained visual encoders (+0.026) and fine-tuning (+0.051) — and none from fusion design. We report four falsified hypotheses alongside these. Our system reaches weighted F_1 0.664 on EGY-MER against a published baseline of 0.519 (single seed; the three-seed mean is 0.654, and 0.669 with the best encoders).

1 Introduction

Multimodal affect recognition rests on the premise that non-verbal channels carry information the words do not. The canonical motivating case is incongruence: an utterance whose literal content is contradicted by its delivery. Cross-modal attention [Tsai et al.,

2019] operationalises this directly, letting one modality condition on another before a decision is made, and has become the default architectural response; it has recently reached Arabic [Ben Cheikhi and Nfaoui, 2026].

This report asks a narrow architectural question and arrives at a broader answer about evaluation. Section 2 describes the two corpora and the protocol. Section 3 introduces *label provenance*, the measurement on which everything else turns, and reports it for both corpora. Section 4 reports the fusion comparison. Section 5 reports the encoder study, which is where the actual gains are. Section 6 lists the hypotheses we tested and rejected. Section 7 states the limitations.

2 Corpora and Protocol

Ar-MuSA [Khaled et al., 2025] comprises 8,700 Egyptian Arabic and MSA clips (8,699 usable) labelled for three-way sentiment with a binary sarcasm flag. Crucially, both are annotated *separately for the transcript, the audio and the frame*, in addition to an overall judgement. The visual stream is a single extracted frame per clip.

EGY-MER [Hussein et al., 2025] is an Egyptian Arabic emotion corpus, obtained from the corpus holders during this work. As delivered it holds 5,471 speaker-diarised segments with media, of which **505 carry a complete annotation** — three annotators, three modalities each, with a documented majority-vote resolution — across 49 source videos and 70 minutes. We use the 484 whose overall label is one of the seven emotions. Unlike Ar-MuSA it ships full video segments, so the visual stream is genuinely temporal; we sample 16 frames per segment.

Protocol. Encoders are frozen and features extracted once, so that the only difference between compared models is the fusion mechanism. Validation is stratified k-fold with identical folds for every model, and predictions are collected out-of-fold so subsets can be analysed without leakage. Because EGY-MER has 484 samples across 7 classes, every EGY-MER number is a mean over three seeds with the standard deviation reported; we treat differences smaller than the seed spread as ties rather than results.

3 Label Provenance

Both corpora annotate each modality separately, so we can ask how much of the overall label a single modality's own annotation already supplies. We call this *label provenance*, and it turns out to determine whether a corpus can evaluate fusion at all.

On Ar-MuSA a predictor with perfect access to the transcript label alone would score macro-F_1 0.9655 on the ostensibly multimodal target; only 304 of 8,699 clips disagree with the text. The sarcasm flag behaves the same way (94.9%). Any fusion comparison on that corpus is therefore a contest over a $\sim$3.5% residue, and a model with no access to audio or vision begins with almost all the available label information. This is a property of the benchmark and applies to every result reported on it, ours included.

EGY-MER inverts this. Its label lives in the *delivery*: text is the weakest channel, audio the strongest, and 40.6% of segments carry an overall label that differs from the text label. This is precisely the regime in which non-verbal channels must carry discriminative information, and it is what the Ar-MuSA result predicted would be required.

Annotation	Ar-MuSA (sentiment)	EGY-MER (emotion)
Transcript	**96.5%**	59.4%
Audio	77.1%	**91.1%**
Frame / video	48.2%	84.4%
Overall label differs from text label	3.5%	**40.6%**

Table 1: How often the overall label equals each modality's own annotation. The ordering inverts between the two corpora. Computed over samples carrying all three modality labels (8,699 and 473) by `experiments/label_provenance.py`.

The prediction is confirmed by model behaviour. Unimodal macro-F_1 inverts to match the annotations (Table 2), and the gain from using all three modalities over the best single one is **three times larger** on EGY-MER (+0.0439) than on Ar-MuSA (+0.0135).

Modality alone	Ar-MuSA	EGY-MER
Text	**0.5458**	0.3404
Audio	0.4184	0.4770
Visual	0.4327	**0.5225**
All three (late fusion)	0.5593	0.5664
Multimodal gain	+0.0135	**+0.0439**

Table 2: Unimodal macro-F_1. The modality ordering inverts with the task, in the same direction as label provenance.

4 Fusion Arrangements

On EGY-MER — the corpus where multimodality demonstrably matters — we compared twelve fusion arrangements under one protocol: plain concatenation; a capacity-matched widening of it; per-modality gating (the strategy the published EGY-MER baseline uses); cross-modal attention anchored on each of the three modalities in turn; an all-pairs arrangement in which every modality attends to every other (following Tsai et al., 2019); full self-attention over the concatenated token sequence; and a hybrid described below.

With the base encoders, plain concatenation scores highest and is the smallest model (891K parameters against 1.1–6.4M). The largest seed standard deviation is 0.021, so the top seven arrangements are statistically indistinguishable; the remaining five are worse. Repeating the benchmark with the best encoders from Section 5 changes the order at the top: a gated hybrid scores 0.608 against concatenation's 0.594, a difference smaller than its own seed deviation (0.017). In neither run does any arrangement beat concatenation by more than seed variation, and the three Tri-CAN variants fall in the lower half both times.

A design flaw in the proposed architecture. The originally specified Tri-CAN feeds its classifier [pool(anchor), pool(anchor's view of A), pool(anchor's view of B)], where every term is pooled over *anchor* positions. All three therefore live in the anchor's token space, and the raw A and B representations never reach the classifier. Tri-CAN is not

late fusion plus attention: it *replaces* the other modalities with the anchor's interrogation of them, and so has strictly less information than the baseline it is meant to beat. This predicts the observed pattern. On incongruent samples every anchor scores below late fusion (-0.047, -0.070 and -0.066 for text, audio and visual anchors), while on congruent samples the text-anchored model is ahead ($+0.036$). The difference-in-differences for the text anchor is -0.082, but its 95% bootstrap interval ($[-0.189, +0.034]$, 208 and 265 samples) includes zero, so we read it as a consistent direction rather than an established effect. On a corpus that is 56% incongruent, the losses dominate.

Restoring the discarded representations (our *hybrid* arrangement, whose input is a strict superset of late fusion's, here with a learned gate) lifts text-anchored attention from -0.0099 to -0.0014 relative to the baseline — from losing to exactly tied. The diagnosis is therefore confirmed, but the repair only recovers parity: the attended view adds nothing the raw vectors did not already carry.

5 Where the Gains Actually Are

Every measurable improvement in this study came from representation, not architecture:

Change	**Gain (macro-F_1)**	**Corpus**
Dialect-matched text encoder (AraBERT-twitter over AraBERTv2)	$+0.054$	Ar-MuSA
Fine-tuning the text encoder	$+0.051$	Ar-MuSA
Reading audio from mid-stack rather than the final layer	$+0.053$	EGY-MER
Expression-pretrained visual encoder over ImageNet Swin	$+0.026$	EGY-MER
Best fusion architecture over plain concatenation	≈ 0	both

Table 3: Representation choices dominate architectural ones by an order of magnitude.

The audio result deserves emphasis because it invalidates a conclusion we had already drawn. In wav2vec2-style models the final layers specialise toward the self-supervised pretraining objective, discarding the slowly-varying paralinguistic content an affect task needs. Reading the final layer cost 0.045–0.053 macro-F_1, and our earlier finding that audio "contributes almost nothing" was measured at the wrong depth.

Fine-tuning is separable from fusion: on Ar-MuSA it improved late fusion and Tri-CAN by exactly $+0.0505$ each, leaving their ordering unchanged. The fusion null is therefore not an artefact of the frozen-feature protocol.

6 Hypotheses We Rejected

We state these because each was predicted in advance and each failed.

1. **Anchor choice.** Since text is EGY-MER's weakest modality, we predicted anchoring attention on audio or vision would help. It did not: text remained the best anchor (audio -0.034, vision -0.009 relative to it).

2. **Multilingual audio encoder.** XLS-R-300m scored *worse* than English-pretrained wav2vec2-base (0.3835 vs 0.4206), so the weak audio signal was not a language mismatch.

3. **Face cropping.** Cropping to the detected face before an expression-pretrained encoder cost -0.0375 despite 98% detection. We attribute this to the corpus being speaker-diarised but not speaker-*localised*: the largest face in a talk-show frame is not necessarily the person speaking, so cropping reliably delivers the wrong face. Active-speaker detection would be required.

4. **Emotion-pretrained audio encoder.** A model fine-tuned for speech emotion recognition scored worse than the generic self-supervised one (0.4578 vs 0.4770); it is tuned on English acted emotion, and that specialisation narrows rather than improves the representation for Egyptian Arabic.

Items 2 and 4 share a lesson worth stating: matching an encoder's *domain* to the data helps, while matching its *task label space* does not.

7 Limitations

1. **Sample size.** EGY-MER's 484 annotated segments across seven classes, with Fear at 12 and Happiness at 18, make macro-F_1 noisy. We report mean $\pm$ SD over seeds and decline to interpret differences below the seed spread.

2. **Baseline comparability.** Our weighted F_1 0.664 (single seed; 0.654 over three) exceeds the published 0.519, but that figure was obtained on 363 samples with different splits; the comparison is indicative, not like-for-like.

3. **Frozen non-verbal encoders.** Only the text encoder was fine-tuned, a compute limitation rather than a design choice.

4. **Annotations, not signal.** "The modalities disagree" means the annotators assigned different labels per stream. It is a good proxy for conflicting evidence, but a proxy.

5. **Grouping.** Folds are stratified by label but not grouped by source video, so some speaker leakage across folds is possible. This affects absolute scores more than the within-study contrasts, which apply equally to every model.

8 Conclusion

Cross-modal attention did not outperform late fusion on either Arabic corpus, under any anchor, arrangement, encoder family, or training protocol we tested. We do not read this as evidence that the mechanism is worthless. We read it as two separate findings.

First, **a benchmark can be structurally incapable of evaluating fusion**, and the standard Arabic multimodal sentiment corpus is: with 96.5% of its label recoverable from the transcript annotation alone, there is no headroom in which mechanisms could be distinguished. Label provenance is cheap to compute, and we suggest reporting it whenever a multimodal benchmark is proposed.

Second, on a corpus that *does* have the headroom, **the binding constraint is representation, not fusion**. Encoder domain, layer depth and fine-tuning moved results by 0.03–0.05 macro-F_1 each; the best of twelve fusion architectures moved them by approximately zero, and the simplest arrangement was also the smallest and the best.

References

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